

भारतीय मानक
Indian Standard

IS 10484 : 2021

पनीर — विशिष्टि

(पहला पुनरीक्षण)

Paneer — Specification

(First Revision)

ICS 67.100.30

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भारतीय मानक ब्यूरो
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June 2021

Price Group 4

Dairy Products and Equipment Sectional Committee, FAD 19

FOREWORD

This Indian Standard (First Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Dairy Products and Equipment Sectional Committee had been approved by the Food and Agriculture Division Council.

Paneer is the indigenous milk product prepared by the combined action of acid coagulation and heat treatment of milk and subsequent drainage of whey. *Paneer* is extensively used as an ingredient for cooking with vegetables in northern India. This standard is expected to help in exercising proper quality control in the manufacturer of *Paneer* of good quality under hygienic conditions.

This Indian Standard was first published in 1983. This revision has been undertaken to harmonize the standard with FSSAI Regulations and the major changes include:

- a) Provision of low fat *paneer* and medium fat *paneer* in the specification.
- b) Alignment of microbiological requirements as per the *Food Safety and Standards (Food Products Standards & Food Additives) Regulations*, 2011.

In the formulation of this standard, due consideration has been given to the *Food Safety and Standards Act*, 2006 and Regulations framed thereunder; the *Legal Metrology Act*, 2009 and Rules framed thereunder and the *Essential Commodities Act*, 1955. However, this standard is subject to restrictions imposed under these, wherever applicable.

The composition of the Committee responsible for formulation of the standard is given in Annex F.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding-off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

PANEER — SPECIFICATION

(First Revision)

1 SCOPE

This standard prescribes the requirements and the methods of sampling and test for *Paneer* including medium fat *Paneer* and low fat *Paneer*.

2 REFERENCES

The following Indian Standards contain provisions which through reference in this text, constitute provision of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below:

<i>IS/ISO No.</i>	<i>Title</i>	<i>IS/ISO No.</i>	<i>Title</i>
		(Part 2) : 1976	Isolation, identification and enumeration of <i>Staphylococcus aureus</i> and <i>faecal streptococci</i> (first revision)
		(Part 3) : 1999	General guidance on methods for the detection of <i>Salmonella</i> (second revision)
		11546 : 2012	Milk and milk products — Guidance on sampling (second revision)
		11917 : 2018	Milk and milk products — Determination of nitrogen content — Kjeldahl principle and crude protein calculation (first revision)
323 : 2009	Rectified spirit for industrial use — Specification (second revision)	13688 : 2020	Packaged pasteurized milk — Specification (second revision)
2311 : 1973	Specification for fat extraction apparatus and milk products (first revision)	14988 (Part 1) : 2020/ISO 11290-1 : 2017	Microbiology of food and feeding stuffs — Horizontal method for detection and enumeration of <i>Listeria monocytogenes</i> : Part 1 Detection method
2491 : 2013	Food hygiene — General principles — Code of practice (third revision)	5402 (Part 1) : 2021/ISO 4833-1 : 2013	Microbiology of the food chain — Horizontal method for the enumeration of microorganisms: Part 1 Colony count at 30 °C by the pour plate technique (third revision)
5401 (Part 1) : 2012/ISO 4832 : 2006	Microbiology — General guidance for enumeration of coliforms: Part 1 Colony count technique (second revision)		
5403 : 1999	Method for yeast and mould count of foodstuffs (first revision)		
5887	Methods for detection of bacteria responsible for food poisoning		
(Part 1) : 1976	Isolation, identification and enumeration of <i>Escherichia coli</i> (first revision)		

3 TERMINOLOGY

For the purpose of this standard, the following definition shall apply.

3.1 Paneer — An important indigenous milk product prepared by the combined action of acid coagulation and heat treatment of any type of milk (as defined in IS 13688) or a combination thereof with or without

IS 10484 : 2021

added milk solids. The phenomenon of precipitation involves the formation of large structural aggregates of proteins in which milk fat and other colloidal and soluble solids are entrained with whey.

4 TYPES

Based on the fat content in the product, *Paneer* shall be of the following types:

- a) *Paneer*;
- b) Medium fat *Paneer*, and
- c) Low fat *Paneer*.

5 REQUIREMENTS

5.1 Colour and Appearance — The product shall have uniform marble white to slightly yellowish off-white colour depending on the type of milk it is prepared from. The product shall be free from any extraneous matter, surface discoloration, adulterants and free moisture.

5.2 Flavour — The product shall have a pleasant odour and mildly acidic, slightly sweet and nutty flavour.

5.3 Body and Texture — The body of product shall neither be too firm nor too soft. The texture of the product shall be compact, smooth, slightly spongy, velvety and free of voids or whey pockets.

NOTE — While masticating, the product should impart a feeling of fibrous texture with modest degree of chewiness. The product should retain its shape while frying.

5.4 All ingredients used shall be clean and in every way fit for human consumption.

5.4.1 The milk may be standardized as required to enable the products to comply with the requirements for composition, body and texture. Before coagulation, milk shall be boiled or heated to sufficiently high temperature for such time that it will result in the complete destruction of pathogenic contaminants.

5.4.2 The coagulants, such as lactic acid, citric acid, malic acid and their sodium and potassium salts, vinegar, glucono delta lactone shall be of food grade and free from toxic substances. Sour whey when used shall be heat-treated to render it microbiologically safe.

5.4.3 The product may contain permitted food additives within the limits as specified under the *Food Safety and Standards (Food Products Standards and Food*

Additives) Regulations, 2011. No extraneous colouring matter shall be added to *Paneer*.

5.5 Hygienic Requirements

The product shall be processed, packed, stored and distributed under strict hygienic conditions as prescribed in IS 2491. It shall be properly packed, stored at low temperature, and distributed under refrigerated/frozen conditions.

5.6 The product shall also comply with the requirements specified in Table 1 and Table 2.

6 PACKING AND MARKING

6.1 Packing — All the materials used for wrapping or packaging of the product shall be food grade and of such a nature as to impart no off-flavour or odour, nor in any other way contaminate the product packed under normal conditions of manufacturer, storage and use.

6.2 Marking

6.2.1 The original pack or the prepared-consumer pack shall be marked as to give the following information:

- a) Name and type of the product and brand name if any;
- b) Name and address of the manufacturer;
- c) Batch or code number;
- d) Date of manufacturing or packing;
- e) Net quantity;
- f) Best before or use by date -----;
- g) Any other requirements under the *Food Safety and Standards (Packaging and Labelling) Regulations*, 2011 and the *Legal Metrology Act*, 2009 and Rules framed thereunder.

6.2.2 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act*, 2016 and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.

7 SAMPLING

Representative samples of the product shall be drawn and tested for conformity to this standard as prescribed in IS 11546.

Table 1 Requirements for *Paneer*, Medium Fat *Paneer* and Low Fat *Paneer*
(Clause 5.6)

Sl No.	Characteristic	Requirement			Method of Test, Ref to
		<i>Paneer</i>	Medium Fat <i>Paneer</i>	Low Fat <i>Paneer</i>	
(1)	(2)	(3)	(4)	(5)	(6)
i)	Moisture, percent by mass, <i>Max</i>	60	60	70	Annex A
ii)	Milk fat, percent by mass (on dry basis)	50 (<i>Min</i>)	More than 20.0 and less than 50.0	20 (<i>Max</i>)	Annex B
iii)	Titrateable acidity (as lactic acid), percent by mass, <i>Max</i>	0.50	0.50	0.50	Annex C
iv)	Protein, percent by mass (on dry basis), <i>Min</i>	30	30	35	IS 11917*
v)	Ash, percent by mass (on dry basis), <i>Max</i>	5.0	5.0	5.0	Annex D

* Protein content is 6.38 multiplied by the total Kjeldahl nitrogen determined.

Table 2 Microbiological Requirements for *Paneer*, Medium Fat *Paneer* and Low Fat *Paneer*
(Clause 5.6)

Sl No.	Characteristic	Requirement				Method of Test, Ref to
		Sampling Plan ¹⁾		Limit (cfu)		
		n	c	m	M	
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Aerobic plate count	5	3	1.5 × 10 ⁵ /g	3.5 × 10 ⁵ /g	Doc. No. FAD 15 (02815)
ii)	Coliform count	5	3	10/g	1 × 10 ² /g	IS 5401 (Part 1)
iii)	<i>Staphylococcus aureus</i> (Coagulase positive)	5	3	10/g	1 × 10 ² /g	IS 5887 (Part 2)
iv)	Yeast and mould count	5	3	50/g	1.5 × 10 ² /g	IS 5403
v)	<i>Escherichia coli</i>	5	0	< 10/g	NA	IS 5887 (Part 1)
vi)	<i>Salmonella</i> sp.	5	0	Absent/25 g	NA	IS 5887 (Part 3)
vii)	<i>Listeria monocytogenes</i>	5	0	Absent/g	NA	IS 14988 (Part 1)
¹⁾ For sampling plan see Annex E.						

¹⁾ For sampling plan see Annex E.

IS 10484 : 2021

ANNEX A

[Table 1, Item (i)]

DETERMINATION OF MOISTURE

A-1 APPARATUS

A-1.1 Flat Bottom Moisture Dish, with Cover-of nickel, stainless steel, porcelain or other suitable material not affected by boiling water, 7 to 8 cm in diameter and not more than 2.5 cm deep, provided with short glass stirring rods having a widened flat end.

A-1.2 Drying Oven, maintained at 102 ± 1 °C.

A-1.3 Desiccator

A-2 PROCEDURE

A-2.1 Weigh accurately about 2 g of the material into a clean dish previously dried and weighed along with a small glass rod. Mix the material-uniformly-with 4 ml of hot distilled water with the help of a small glass rod. Wash off the particles of material adhering to the glass rod by pouring an additional 1 ml hot distilled water.

Heat the dish containing the material after uncovering in the electric oven maintained at 102 ± 1 °C for about 4 h. Cool the dish in the desiccator and weigh with the cover on. Repeat the process of drying, cooling and weighing at 30 minutes interval until the difference between the two consecutive weighings is less than one milligram. Record the lowest weight.

A-3 CALCULATION

A-3.1 Moisture percent by mass = $\frac{100 (M_1 - M_2)}{M_1 - M}$

where

M_1 = mass in g, of the dish with material before drying;

M_2 = mass in g, of the dish with the material after drying; and

M = mass in g, of the empty dish.

ANNEX B

[Table 1, Item (ii)]

DETERMINATION OF FAT

B-1 APPARATUS

B-1.1 Fat Extraction Apparatus, as described in IS 2311.

B-1.2 Electric Oven, well-ventilated and maintained at 100 ± 1 °C.

B-2 REAGENTS

B-2.1 Hydrochloric Acid, specific gravity 1.125.

B-2.2 Ethyl Alcohol (distilled rectified spirit).

B-2.3 Diethyl Ether, Sp gr 0.720, free from peroxide. It may be maintained free from peroxide by adding wet zinc foil (approximately 80 cm² per litre, cut in strips long enough to reach at least half way up the container) that has been completely immersed in dilute acidified copper sulphate solution for one minute and subsequently washed with water.

B-2.4 Light Petroleum Ether, boiling range 40 to 60 °C, recently distilled.

B-3 PROCEDURE

B-3.1 Extraction

Weigh accurately about 1 gm of sample into a clean dry 50-ml beaker. Add 8 ml of hot distilled water and then 3 ml of HCl solution. Warm and swirl gently-the mixture till *Paneer* is dissolved completely. Cool the mixture. With 10 ml of ethyl alcohol, transfer the contents to the Mojonnier fat extraction apparatus. Mix well.

B-3.1.1 Add 25 ml of diethyl ether through the beaker used for weighing the sample. Close the extraction flask and shake vigorously and invert repeatedly for 1 min. It is essential that the cork (or stopper) be wetted before each insertion and washed with solvent during each removal.

B-3.1.2 Open the flask and add 25 ml of light petroleum. Close the flask and shake vigorously and invert repeatedly for 1 min. Allow the apparatus to stand until the upper layer has become clear and is distinctly separated from the aqueous layer. Alternately, perform the separation by the use of suitable centrifuge.

B-3.1.3 Carefully transfer as much as possible of the supernatant layer by decantation into suitable flask containing 2 glass beads. Wash the outside of the neck of the flask and cork or stopper with mixed solvent, collecting the rinsings in the flask. Make second and third extractions by repeating the procedure using 15 ml each of diethyl ether and light petroleum ether.

B-3.2 Distil carefully the solvent from the flask collected under **B-3.1.3** and completely remove the solvent. Wipe the flask and dry the residual fat in the oven at 98 to 100 °C for one hour, taking precaution to remove all traces of volatile solvent and cooling the flask to room temperature in a desiccator with efficient desiccant.

B-3.2.1 Extract the fat from the flask with successive 15 ml of light petroleum. After the first addition, the flask should be warmed and the solvent swirled round the sides until all the fat appears to be in solution. Allow any sediment to settle and carefully decant the solution without disturbing the sediment through

lightly packed cotton wool in a small funnel, to a weighed flask containing two glass beads. Repeat at least twice. Finally rinse the neck of the flask with light petroleum and after allowing the flask to stand, decant. As described above, distil off the solvents, heat the flask in the oven, allow to cool and weigh. Repeat heating in oven, cooling and weighing until successive weighings do not show a loss in mass by more than one milligram.

B-3.3 Simultaneously when the above procedure is carried out, make a blank determination with 1 to 2 ml of water in place of the sample. Use a similar extraction apparatus, the same reagents and the same technique throughout. The difference in mass before and after the petroleum extractions, after correcting for the blank, is the mass of the fat contained in the mass of the sample of *Paneer* taken.

B-3.4 Calculation

Calculate the percentage of fat by mass in *Paneer*. The maximum deviation between duplicate determinations shall not, exceed 0.2 percent of fat.

ANNEX C

[Table 1, Item (iii)]

DETERMINATION OF TITRATABLE ACIDITY

C-1 APPARATUS

C-1.1 Burette, with soda-lime guard tube.

C-1.2 Porcelain Dishes, white, hemispherical of approximately 60 ml capacity.

C-1.3 Pipettes, to deliver 10 ml and 1 ml.

C-1.4 Pestle and Mortar

C-2 REAGENTS

C-2.1 Standard Sodium Hydroxide Solution 0.1 N — Prepare a concentrated stock solution of sodium hydroxide by dissolving equal parts of sodium hydroxide (sticks or pellets) and water in a flask. Tightly stopper the flask with a rubber bung and allow any insoluble sodium carbonate to settle down for 3 to 4 days. Use the clear supernatant liquid for preparing the standard 0.1 N solution. About 8 ml of stock solution is required per litre of distilled water.

C-2.2 Phenolphthalein Indicator Solution — Dissolve one gram of phenolphthalein in 100 ml of rectified spirit (see IS 323). Add 0.1 N sodium hydroxide solution until one drop gives a faint pink coloration. Dilute with distilled water to 200 ml.

C-2.3 Standard Hydrochloric Acid — 0.1 N.

C-3 PROCEDURE

C-3.1 Weigh accurately about 2 g of *Paneer* into a porcelain dish. Add 3 ml of boiling distilled water and render the sample into a fine paste using a pestle and mortar. Dilute by another 17 ml of boiling distilled water washing off the adherents from the pestle. Cool to room temperature. Add 10 ml of 0.1 N sodium hydroxide. Add 1 ml of 0.5 percent phenolphthalein indicator and titrate against 0.1 N hydrochloric acid till the pink colour disappears. Stir vigorously throughout.

C-4 CALCULATION

C-4.1 Titratable acidity (as lactic acid), percent by

$$\text{mass} = \frac{10 - V}{M} \times 0.9$$

where

V = volume of 0.1 N hydrochloric acid required for titration, and

M = mass in g, of sample of product.

IS 10484 : 2021

ANNEX D

[Table 1, Item (v)]

DETERMINATION OF ASH

D-1 APPARATUS

D-1.1 Flat Bottom Dish, of stainless steel, porcelain, silica or platinum.

D-1.2 Muffle Furnace, maintained at 550 ± 20 °C.

D-1.3 Desiccator

D-2 PROCEDURE

D-2.1 Weigh accurately about 5 g of the material in the tared dish, previously dried in an air-oven. Heat the dish gently on a flame at first and then strongly in muffle furnace at 550 ± 20 °C till grey ash results. Cool the dish in a desiccator and weigh. Heat the dish again

at 550 ± 20 °C for 30 min. Cool the dish in a desiccator and weigh. Repeat this process of heating for 30 min, cooling and weighing until the difference between two successive weighings is less than one milligram. Record the lowest mass.

D-3 CALCULATION

D-3.1 Total ash, percent by mass = $\frac{100 (M_2 - M)}{M_1 - M}$

where

M_2 = mass in g, of the dish with the ash;

M = mass in g, of the empty dish; and

M_1 = mass in g, of the dish with the material taken for the test.

ANNEX E

[Table 2]

SAMPLING PLAN FOR MICROBIOLOGICAL REQUIREMENTS

E-1 SAMPLING PLAN FOR MICROBIOLOGICAL REQUIREMENTS

The terms n, c, m and M used in this standard have the following meaning:

n = Number of units comprising a sample.

c = Maximum allowable number of units having microbiological counts above m for 2-class sampling plan and between m and M for 3-class sampling plan.

m = Microbiological limit that separates unsatisfactory from satisfactory in a 2-class sampling plan or acceptable from satisfactory in a 3-class sampling plan.

M = Microbiological limit that separates unsatisfactory from satisfactory in a 3-class sampling plan.

E-2 INTERPRETATION OF RESULTS

2-Class Sampling Plan (where n, c and m are specified)	3-Class Sampling Plan (where n, c, m and M are specified)
1. Satisfactory, if all the values observed are $\leq m$ 2. Unsatisfactory, if one or more of the values observed are $> m$ or more than c values are $> m$	1. Satisfactory, if all the values observed are $< m$ 2. Acceptable, if a maximum of c values are between m and M and the rest of the values are observed as $\leq m$ 3. Unsatisfactory, if one or more of the values observed are $> M$ or more than c values are $> m$

ANNEX F

(Foreword)

COMMITTEE COMPOSITION

Dairy Products and Equipment Sectional committee, FAD 19

<i>Organization</i>	<i>Representative(s)</i>
National Dairy Research Institute, Karnal	DR MANMOHAN SINGH CHAUHAN (Chairman)
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Jupitor Glass Works, New Delhi	MR KARAN NANGIA MR AMREEK SINGH PURI (<i>Alternate</i>)

IS 10484 : 2021

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This Indian Standard has been developed from Doc No.: FAD 19 (16191).

Amendments Issued Since Publication

Amend No.	Date of Issue	Text Affected

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Published by BIS, New Delhi